

Complexity analysis of the RTIGER core optimizations

A per-step big- $\mathcal{O}$ accounting of the `optimize-julia-core` fork

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Abstract

This note derives, step by step, the asymptotic cost of the rigid-HMM (rHMM) Baum–Welch fit that RTIGER performs, and the big- $\mathcal{O}$ improvement contributed by each optimization in the `optimize-julia-core` fork. Equation numbers refer to the supplementary text of Campos-Martin *et al.* (2023).¹ Every optimization preserves the arithmetic of the model (Sections 2–4 of the supplement); only the cost of evaluating it changes. Two of the optimizations remove an asymptotic factor (the rigidity window r , and the marker count T in the emission line search), one removes a memory factor (the sample count N), and two are constant-factor (allocation) wins that leave the complexity class unchanged.

1 Notation and cost model

symbol	meaning
T	markers (positions) in one observation chain
C	number of chains (samples $\times$ chromosomes); $T_{\text{tot}} = \sum_c T_c$
N	number of samples (so C and T_{tot} grow $\propto N$)
$ S = s$	number of hidden states (= 3: paternal / heterozygous / maternal)
r	rigidity; augmented rigid state space $S' = S \times \{1, \dots, r\}$, $ S' = sr$
D	number of <i>distinct</i> emission pairs (k, n) in a chain
E	objective evaluations per state in one emission-M-step line search

Table 1: Cost parameters. For low-coverage genotyping data $D \ll T$: the read depth n is small, so the distinct (k, n) pairs number a few dozen regardless of T .

The emission is a Beta–Binomial (supplement §3.3), $\psi_s(k; n, a_s, b_s) = \binom{n}{k} B(k+a_s, n-k+b_s)/B(a_s, b_s)$, whose evaluation costs a handful of `lgamma` calls — the dominant scalar op of the fit. A full fit is

$$\text{cost} = (\text{\#EM iterations}) \times \sum_{c=1}^C (\text{E-step}_c + \text{M-step}_c),$$

so we analyse the per-iteration, per-chain cost and then note how it aggregates.

¹Campos-Martin R, Schmickler S, Goel M, Schneeberger K, Tresch A. *Reliable genotyping of recombinant genomes using a robust hidden Markov model*. Plant Physiology 2023;192(2):821–836. doi:10.1093/plphys/kiad191; PMC10231367. <https://doi.org/10.1093/plphys/kiad191>

2 The auxiliary window product Ψ_k^t (productpsi)

The rigid recursions repeatedly need the windowed emission product (supplement Eq. 21)

$$\Psi_k^t = \prod_{u=\max(t-r+1,1)}^t \psi_k(o_u), \quad 1 \leq t \leq T. \quad (21)$$

Computing each of the $T \cdot s$ values by re-multiplying its length- r window costs $\mathcal{O}(Tsr)$. In log space $\log \Psi_k^t = \sum_u \log \psi_k(o_u)$ is a fixed-width sliding sum: maintain a running window, add the entering term and drop the leaving term in $\mathcal{O}(1)$ per step.

$$\boxed{\mathcal{O}(Tsr) \rightarrow \mathcal{O}(Ts)} \quad (\text{removes the rigidity factor } r; \text{ commit d725651}).$$

At the production rigidity $r = 250$ this is a $\sim r$ -fold reduction of the Ψ term.

3 Emission log-pdf memoization (getlogpsi)

Each forward, backward and Viterbi step evaluates $\psi_s(o_t)$ at every position (Eqs. 22–27, 75). Done literally, that is $\mathcal{O}(Ts)$ Beta–Binomial evaluations per iteration — each one several `lgamma` calls. But $\psi_s(o_t)$ depends on $o_t = (k_t, n_t)$ only, and there are only D distinct pairs. Memoizing $\log \psi_s$ over the distinct (k, n) replaces the expensive evaluations with cheap table lookups:

$$\boxed{\mathcal{O}(Ts) \text{ Beta–Binomial evals} \rightarrow \mathcal{O}(Ds) \text{ evals} + \mathcal{O}(Ts) \text{ lookups}}$$

i.e. the costly part shrinks by the factor T/D (commit d725651).

4 Emission M-step line search (the dominant cost)

The Beta–Binomial parameters are fit by a 1-D line search over τ_s that maximizes (supplement Eqs. 38, 43)

$$Q(\tau_s) \propto \sum_{c=1}^C \sum_{t=1}^{T_c} \gamma_s^{t,c} \log \psi_s(k_t^c; n_t^c, a_s = m_s \tau_s, b_s = (1 - m_s) \tau_s). \quad (43)$$

Original. Each of the E line-search evaluations rebuilds a Beta–Binomial object and sums $\log \psi_s$ over all T_{tot} markers: $\mathcal{O}(EsT_{\text{tot}})$ Beta–Binomial constructions per iteration. The supplement notes the forward/backward/Viterbi kernels are the “time-critical” parts; in practice this repeated full-marker objective is what dominates — empirically $\sim 99.97\%$ of upstream runtime.

Optimized. Because the sum only depends on the data through the γ -weighted counts at each distinct pair, regroup it once:

$$Q(\tau_s) = \sum_{(k,n) \in \mathcal{D}} \underbrace{\left(\sum_{t: (k_t, n_t) = (k,n)} \gamma_s^t \right)}_{G_s(k,n) \text{ (pre-summed once)}} \log \psi_s(k; n, a_s, b_s). \quad (1)$$

The weights $G_s(k, n)$ are accumulated in one $\mathcal{O}(sT_{\text{tot}})$ pass; each line-search evaluation then costs $\mathcal{O}(sD)$:

$$\boxed{\mathcal{O}(EsT_{\text{tot}}) \rightarrow \mathcal{O}(sT_{\text{tot}} + EsD)}$$

Since $ED \ll T_{\text{tot}}$, the M-step collapses from “ E passes over all markers, rebuilding objects” to “one binning pass” — a measured $\sim 1500\times$ at production scale (commit 16b8a65). After this, the M-step is no longer the bottleneck.

5 Forward / backward and Viterbi (constant-factor)

The rigid forward (Eqs. 22–23), backward (Eqs. 26–27) and Viterbi (Eq. 75)

$$\varphi_k^t = \max_{j \in S} \begin{cases} \psi_k(o_t) a_{kk} \varphi_k^{t-1} & j = k \\ \Psi_k^t a_{jk} \varphi_j^{t-r} & j \neq k \end{cases} \quad (75)$$

each take a max/sum over s predecessors at every position, giving $\mathcal{O}(T s^2)$ per chain (the dwell-counter shifts for $k < r$ are $\mathcal{O}(1)$ copies once Ψ is available). The optimizations here — in-place scalar log-sum-exp (Eq. 80) and an allocation-free in-place arg-max — do *not* change the class:

$$\mathcal{O}(T s^2) \longrightarrow \mathcal{O}(T s^2), \quad \text{constant-factor: } \sim 5 \times (\text{fwd/bwd, commit 44ed85b}), \sim 12 \times (\text{Viterbi, 6cf85ca}),$$

with $\sim 90 \times$ fewer allocations in the forward/backward kernel. Once the emission M-step is fixed (§4), these $\mathcal{O}(T s^2)$ kernels are what the per-iteration cost reduces to — hence *linear in T* .

6 Memory: streaming the M-step (eb79933)

The M-step needs the data only through pooled sufficient statistics that are *sums* over chains (Eqs. 40–42): $\sum_c \zeta$, $\sum_c \gamma^1$, and the γ -weighted counts $G_s(k, n)$.

Original. `EM()` held, for *all* C chains at once, the per-chain arrays ζ ($T \times s^2$), γ ($s \times T$) and α, β, ψ ($sr \times T$), pooling only at the end:

$$\text{peak} = \mathcal{O}\left(\sum_c T_c (s^2 + sr)\right) = \mathcal{O}(N T (s^2 + sr)) \quad (\text{linear in } N).$$

Streaming. Fold each chain’s contribution into the running sums inside the E-step loop and discard its arrays immediately (the public `@Probabilities` slot, which nothing reads, is no longer filled):

$$\boxed{\mathcal{O}(N T (s^2 + sr)) \longrightarrow \mathcal{O}(T_{\max} (s^2 + sr)) + \mathcal{O}(T_{\text{tot}})}$$

peak scratch becomes one chain’s worth, *independent of N* ; the only N -linear residual is the input count matrices that must be held. Measured: ~ 33 GB projected $\rightarrow \sim 3.6$ GB at $1400 \times 50k$.

7 Summary

8 Reconciliation with the measured scaling

A marker-scaling head-to-head (one shared 109,703-locus panel, decimated by odd index to five sizes, both cores run to convergence) gives empirical per-iteration exponents

$$\text{original} \approx \text{markers}^{2.1}, \quad \text{optimized} \approx \text{markers}^{1.1}.$$

These are consistent with the analysis:

- **Optimized.** With the M-step reduced to a binning pass plus an $\mathcal{O}(EsD)$ search (D nearly flat), the per-iteration cost is dominated by the $\mathcal{O}(T s^2)$ forward/backward/Viterbi kernels — *linear in T* , matching the measured ~ 1.1 exponent.

optimization	before (per iter)	after (per iter)	gain
Ψ window (<code>productpsi</code>)	$\mathcal{O}(Tsr)$	$\mathcal{O}(Ts)$	drop r
emission log-pdf (<code>getlogpsi</code>)	$\mathcal{O}(Ts)$ evals	$\mathcal{O}(Ds)$ evals	T/D
emission M-step (<code>16b8a65</code>)	$\mathcal{O}(EsT_{\text{tot}})$	$\mathcal{O}(sT_{\text{tot}} + EsD)$	$\sim 1500\times$
forward/backward (<code>44ed85b</code>)	$\mathcal{O}(Ts^2)$	$\mathcal{O}(Ts^2)$	$\sim 5\times$ (const.)
Viterbi (<code>6cf85ca</code>)	$\mathcal{O}(Ts^2)$	$\mathcal{O}(Ts^2)$	$\sim 12\times$ (const.)
memory, streaming (<code>eb79933</code>)	$\mathcal{O}(NT(s^2+sr))$	$\mathcal{O}(T_{\text{max}}(s^2+sr))$	drop N

Table 2: Per-EM-iteration time complexity (top) and peak-memory complexity (bottom). $s = 3$ is constant; the asymptotically meaningful factors are T , r , D , E and N .

- **Original.** The per-iteration cost is dominated by the emission line search $\mathcal{O}(EsT_{\text{tot}})$, but E (the number of `Optim` objective evaluations) is data-dependent and grows with scale, and the Beta-Binomial object is rebuilt per marker. The result is an empirically super-linear, erratic per-iteration cost ($\sim \text{markers}^{2.1}$, with iteration-to-iteration spikes) rather than a clean $\mathcal{O}(T)$ — exactly the behaviour the grouping in §4 removes.

The net effect: per-iteration cost goes from empirically $\sim$ quadratic and erratic to provably linear ($\mathcal{O}(Ts^2)$), and the optimized-vs-original speed-up therefore *widens* with problem size (from $\sim 34\times$ at $\sim 7\text{k}$ markers/sample to $\sim 610\times$ at $\sim 110\text{k}$) while every fit remains numerically identical (matching iteration counts, identical Viterbi paths, parameters agreeing to $\leq 3 \times 10^{-6}$).